POWER TORSION MODULES BECOME INVARIANT UNDER BASE CHANGE EXACTLY WHEN THE QUOTIENTS BY THE IDEAL AGREE

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Editorial note. The counted statement begins by assuming that the pair satisfies the equivalent conditions of the preceding lemma; that lemma's own statement is delivered as the context span of this card, so the hypothesis can be read and its label resolves inside the compiled document without a navigation stub.

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Allowed standard prerequisites (editorial).

- A module annihilated by the ideal is a module over the quotient, and tensoring it over the quotient with the base changed quotient gives back the module when the quotients agree. used at proof lines 24279-24284. Proves the base case of the induction, which is the heart of the equivalence.
- Flatness of the ring map makes the tensor product with it exact, so a short exact sequence of modules gives a short exact sequence after tensoring. used at proof lines 24288-24293. Carries the induction step from the submodule and the quotient to the whole module.
- A power torsion module is the union of the submodules annihilated by the successive powers of the ideal, and tensor products commute with colimits. used at proof lines 24294-24297. Passes from the modules annihilated by a fixed power to an arbitrary power torsion module.

Lemma 0.2. *Let $\varphi : R \to S$ be a ring map. Let $I \subset R$ be an ideal. The following are equivalent*

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(1) *φ is flat and $R/I \to S/IS$ is faithfully flat,*
(2) *φ is flat, and the map $\mathrm{Spec}(S/IS) \to \mathrm{Spec}(R/I)$ is surjective.*
(3) *φ is flat, and the base change functor $M \mapsto M \otimes_R S$ is faithful on modules annihilated by I, and*
(4) *φ is flat, and the base change functor $M \mapsto M \otimes_R S$ is faithful on I-power torsion modules.*

Lemma 0.3. *Assume $(\varphi : R \to S, I)$ satisfies the equivalent conditions of Lemma 0.2. The following are equivalent*

(1) *for any I-power torsion module M, the natural map $M \to M \otimes_R S$ is an isomorphism, and*
(2) *$R/I \to S/IS$ is an isomorphism.*

Proof. The implication (1) $\Rightarrow$ (2) is immediate. Assume (2). First assume that M is annihilated by I. In this case, M is an R/I-module. Hence, we have an isomorphism

$$M \otimes_R S = M \otimes_{R/I} S/IS = M \otimes_{R/I} R/I = M$$

proving the claim. Next we prove by induction that $M \to M \otimes_R S$ is an isomorphism for any module M is annihilated by I^n. Assume the induction hypothesis holds for n and assume M is annihilated by I^{n+1}. Then we have a short exact sequence

$$0 \to I^n M \to M \to M/I^n M \to 0$$

and as $R \to S$ is flat this gives rise to a short exact sequence

$$0 \to I^n M \otimes_R S \to M \otimes_R S \to M/I^n M \otimes_R S \to 0$$

Using that the canonical map is an isomorphism for $M' = I^n M$ and $M'' = M/I^n M$ (by induction hypothesis) we conclude the same thing is true for M. Finally, suppose that M is a general I-power torsion module. Then $M = \bigcup M_n$ where M_n is annihilated by I^n and we conclude using that tensor products commute with colimits. $\square$